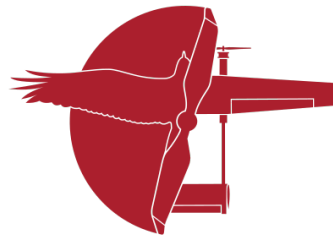

University of Bristol

Rescue mission using smart drone – Flying Aid

By

TEAM GREENCO



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Github link: <https://github.com/argi4s/Group-Project-GreenCo>

AI use declaration: AI utilised to support code development and to facilitate some translations

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EXECUTIVE SUMMARY

This report presents the design, development and evaluation of an automated drone-based Search and Rescue (SAR) system, developed by Team GreenCo at the University of Bristol. The system was built upon a Hexsoon EDU-450 quadrotor platform, augmented with a Raspberry Pi 5 companion computer, an Orange Cube flight controller and a custom vision pipeline to detect and localise a casualty over a defined search area at Fenswood Farm.

The system architecture follows a modular, segmented mission design, decomposed into a Finite State Machine (FSM) controlling discrete mission phases: takeoff, area-of-interest transit, lawnmower search, PLB-guided focused search, visual alignment, descent and payload delivery. A custom Automation Ground Control Station (AGCS) provided the operator interface, coordinating with Mission Planner as the safety-critical fallback. Communication between system components was implemented over UDP to meet the low-latency demands of the perception-control loop.

A YOLO-based vision module deployed via OpenVINO on the Raspberry Pi provided real-time humanoid detection, with normalised position offsets used to drive the ALIGN state for target-centred descent. An altitude-aware lawnmower path planner, scaled to camera footprint, ensured systematic area coverage while respecting geofenced boundaries and SSSI exclusion zones.

The project was evaluated against twelve system requirements across flying, search and payload autonomy domains, achieving Level 2 flying autonomy (F2), Level 2 search autonomy (S2) and Level 1 payload autonomy (P1) outside of simulation. Key requirements including flight area containment, SSSI avoidance, maximum altitude, PLB response and designated pilot roles were met. Autonomous landing proximity and positional reporting were not achieved within the project timeline.

Four trial flights at Fenswood Farm validated the system progressively, culminating in successful AOI navigation following resolution of FSM logic and MAVLink message filtering issues encountered during Flight Day 3. The report concludes with an evaluation of technical performance, environmental and societal impact and future development directions, including full ALIGN validation, terrain-aware altitude control and improved communication security.

I. INTRODUCTION

A. Context and Background

Search and Rescue (SAR) missions have, in recent years, become increasingly important across a variety of environments, including land, maritime and mountainous regions. The concept of the three “Ds” (dull, dirty and dangerous) provides a useful framework for understanding why robotic systems are deployed in such tasks. In this context, the “dangerous” aspect is the main driver, as rescue operations often take place in hazardous, unstructured and difficult-to-access locations.

A key challenge in SAR operations is the detection and localisation of individuals over large areas, often at a distance and under conditions of partial or full visual obstruction (e.g., vegetation, terrain, or debris). In addition, defining efficient search and tracking patterns remains a complex problem, particularly in dynamic and uncertain environments.

Traditional SAR methods, such as helicopter deployment or large-scale ground searches, are a high-cost but high-value solution. They require highly trained personnel and incur significant operational costs. In contrast, drone-based solutions offer a more cost-effective and flexible alternative, requiring less expensive equipment and lower training barriers (e.g., a drone operator versus a helicopter pilot), while still enabling rapid area coverage.

However, drone-assisted SAR systems also have limitations. While conventional methods can deploy trained responders directly to the scene, drones are constrained by payload capacity and can only deliver limited aid. As such, they are best viewed as complementary tools or situation-dependent, particularly valuable in the early stages of search and localisation.

Furthermore, the use of Personal Locator Beacons (PLBs) can significantly improve efficiency, as their signals can be effectively detected by aerial platforms such as drones, enabling faster and more precise localisation compared to traditional approaches.

Overall, this project explores an automated drone-based solution as a practical and scalable enhancement to modern SAR operations.

B. Company Description

The assigned company name was GreenCo. Embracing this identity proved to be an outstanding choice, as the colour green is commonly associated with safety, environmental friendliness and sustainability. Our philosophy was directly inspired by these values.

C. Member Bios, Roles and Contributions

The team responsible for developing this rescue operation comes from a diverse academic background. As a result, tasks were allocated based

on each member’s individual skillset. The team consisted of:

- A developer responsible for the Finite State Machine (FSM), module integration and simulation
- A Computer Vision module developer
- A path planning developer
- A geofencing and risk assessment specialist, also responsible for the report and presentation
- A coordinator responsible for team management, system oversight and acting as the company pilot

The project timeline and task allocation are summarised in the Gantt chart (see Appendix, Fig. 14), alongside a risk assessment matrix documenting the identified management risks and their mitigations (Fig. 15).

D. Mission Description

This project aims to conduct a successful rescue mission in a crucial situation using Hexsoon EDU 450, such as a landslide or earthquake, or to find a lost person. To scan a large area within a short span of time and deliver a first-aid kit in case there is an injured person who is at an unreachable distance, or to deliver a food package during a time of flood. Planning a rescue mission has a number of benefits. As part of the project, we have been assigned to initiate a rescue mission for a hiker who is injured during hiking. This project aims to achieve the following tasks:

- To find a hiker traversing the Fenswood Wilderness with a given description of his height, clothing colour and equipment within a defined area as the hiker activates the PLB, helping the mission to focus on a particular Area of Interest (AOI).
- Upon PLB activation, the team is expected to conduct a search mission over the flight area shown in Fig. 1. Given a time limit of 5 to 15 minutes, look for the hiker with the given description.
- Required to fly within the flight area and avoid entering the geofence region and Site of Specific Scientific Interest (SSSI).
- Upon successful detection of the hiker, the drone is expected to descend and land within 5 to 10 meters from the casualty, deploy the first aid kit, then return to the takeoff location (TOL).

II. DESIGN RATIONALE

A. Coverage Method Selection

1) Search Patterns Considered

Classical SAR guidance recommends selecting a search pattern according to datum confidence, search-area size, geographical context, and available resources [1]. For this project, the main candidate

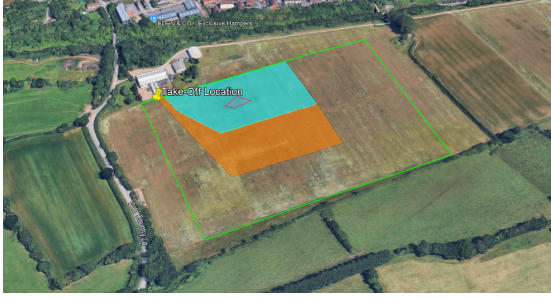


Fig. 1: The Fenswood area with geofence data

patterns considered were sector search, expanding square search, parallel track search, and creeping line search [1]. Sector search is intended for a small area with a well-established datum and can offer a high probability of detection, although its efficiency decreases when the search legs become too long relative to the sweep width [1]. Expanding square search is also suitable when the datum is reasonably well established and little time has elapsed, but it generally offers a lower probability of detection than sector search [1]. For larger areas where the target position is not well established, parallel track search is more appropriate, while creeping line search is better suited to cases in which the target is considered more likely to lie toward one end of the search area [1]. The SAR guide also notes that parallel track and creeping line patterns are versatile and among the most commonly used search methods [1].

2) Selected Approach Justification

Chosen approach: lawnmower. The final system adopts a lawnmower search pattern, which corresponds most closely to a parallel-track search in SAR terminology. This approach was selected because it is straightforward to implement, easy to verify visually during testing, and well-suited to systematic coverage of a bounded search polygon. In practical terms, structured coverage depends on maintaining consistent track spacing between successive search legs, since search effectiveness is linked to the relationship between sweep width and track spacing. A regular back-and-forth pattern, therefore, provides a clear and defensible basis for area coverage in the initial search phase.

B. Segmented Autonomous Mission

Segmented autonomous mission refers to the separation of major elements in the rescue mission into steps. This choice means actions such as 'take off' and travelling to the Area of Interest (AOI) are not a single smooth motion. Each requires a unique input from the Graphical User Interface (GUI).

For this project a segmented mission allowed a modular design approach where each member contributes a module independently and until its integration other modules can still be tested. For

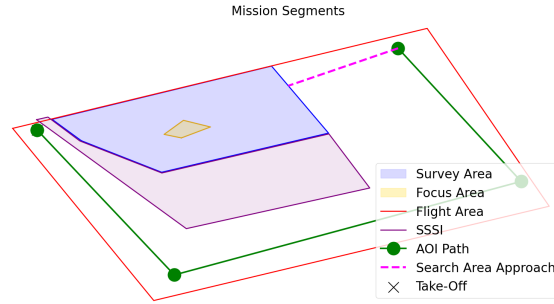


Fig. 2: Early mission render indicating AOI travel and travel into search area

instance, the ALIGN routine did not inhibit test and execution of the 'take off', 'AOI' and 'lawn' modules with it absent. Each segment is in turn allocated a state in the Finite State Machine.

Fig. 2 indicates the AOI path in green, implemented using hard coded waypoints that travel around the SSSI.

In a safety context, this also enables the team to interrupt the mission and visually identify the end of each action. As a continuous flow makes it challenging to observe at what part of flight the drone is entering. Raises flight team confidence when anticipating state changes. The detriment of this is that in a real time sensitive mission, these momentary stops and unstable flow may increase the time it takes to carry out the rescue and every moment counts for a victim in a vulnerable state.

C. Autonomy Levels

Autonomy describes the agency of a system to make its own decisions and carry out given or derived tasks. The Society of Automotive Engineers (SAE) have defined international standards used by the industry that describe 6 levels of Autonomy [2].

- **Level 0:** No Driving Automation - warnings and indicators but no automation
- **Level 1:** Driver Assistance - braking or steering automation features
- **Level 2:** Partial Driving Automation - simultaneous braking and steering automation features
- **Level 3:** Conditional Driving Automation - select fully autonomous tasks, e.g. Parking
- **Level 4:** High Driving Automation - Fully autonomous driving under limited conditions
- **Level 5:** Full Driving Automation - Fully autonomous driving under all conditions

As described in [2], the operator is actively in control at levels 0 to 2 while automation features are engaged. For levels 3 and above the automation system operates without the operator, though at level 3 the feature may require the operator's input. The autonomy feature set for this project was designed to align with the SAE framework, applied across three domains: Flying Autonomy (F), Search Autonomy

(S) and Payload Autonomy (P), each represented by 6 levels from 0 to 5, as defined in Figure 16.

1) Minimum viable product

- $F = 1, S = 2, P = 1$

The minimum viable product reflects using automation assistively for an otherwise autonomous mission. The performance at levels from 0 to 2 can be observed using a standard flight controller paired with Ground Control Station on most equipped drones. Therefore, as a minimum the company intends to demonstrate this assistive capability and otherwise conduct a manual mission.

1F indicates the use of a basic STABILIZE or ALT HOLD mode to fly the drone, between points of interest or to the search area. This requires manually steering within the flight area with the assistance of Mission Planner [3] to remain within those bounds. Conducting the mission Level 0 may be possible but will require a significant effort and skill from the Company Pilot, increasing the risk of a potential crash or erroneous movement.

2S suggests that the visual detection will be performed autonomously with human confirmation. This has been selected as during flight as the drone banks and moves manually tracking the visual target can become challenging for the human but any detection algorithm in a semi-structured environment has the chance of indicating false positives. These working alone in an autonomous system could initiate actions that spend time and put the subject awaiting rescue at threat therefore human confirmation is required.

1P describes the use of a wireless signal to trigger the actuator for payload deployment. The alternative at level 0 is for the injured party to manually disengage the payload. This will severely limit the value of the automated drone operation as it will expect the operator to be in a fit state to learn and operate an unfamiliar mechanism while in the need of medical help.

2) Maximum viable product

- $F = 3, S = 4, P = 3$

The maximum viable product was constrained to require a human in the loop. Civil Aviation Authority (CAA) regulations do not permit Beyond Visual Line of Sight (BVLOS) operations without explicit permission [4]. Even under direct observation, the level of risk must be kept As Low As Reasonably Practicable (ALARP). With only four trial flights available, the team prioritised safety and reliability over scope. The operational environment requires further consideration, as Fenswood Farm presents a semi-structured setting where conditions on any given trial day are subject to change by weather effects or terrain foliage.

D. Geofence

1) Definition

A geofence is a virtual geographic area that is enabled to protect civilian airspace, airport-controlled airspace, and restricted areas such as military, scientific research, and protected environmental areas. As discussed in [5]. In the case of automated drone operations, the geofence feature can be enabled using the mission planner. Inclusion and exclusion zones enhance the safety and reliability of the operation. Setting maximum and minimum altitudes will try to prevent exceeding these limits, automatically declaring a fence breach in all modes if they are exceeded.

2) Regulatory Justification

i. CAA rules

Under the regulatory framework established by the CAA, unmanned aerial vehicles (UAVs) are categorised according to their weight and operational capabilities. These classifications determine the applicable rules and restrictions for each category, ensuring safe operation and adherence to UK aviation safety standards. The following Table I has been taken from the official website [6].

Category	UK class mark	Weight
Open Over People (A1)	UK0	Less than 250g
Open Over People (A1)	UK1	Less than 900g
Open Near People (A2)	UK2	Less than 4kg ($< 2\text{kg}$ rules apply)
Open Far From People (A3)	UK0, UK1, UK2, UK3, UK4	Less than 25kg
Specific Category	UK5, UK6	Less than 25kg

TABLE I: UK drone classification (CAA)

The calculated weight of the drone and specified components is as follows:

- Hexsoon EDU-450: 870g and maximum payload: 960g (without battery)
- Here 3+: 51.8g
- FrSky TWmini: 2.2g
- Raspberry Pi 5: 4.75g
- Raspberry Pi global shutter camera: 34g
- Pi camera lens: 53g
- SD card: 4.54g
- Actuator: 57g

DRONE WEIGHT CLASSIFICATION

After weighing the drone on a scale, the total weight was calculated as shown in Table II: At a total MTOM of 2,033 g ($\approx 2.03\text{ kg}$), the drone falls within the **UK2 class weight range** (under 4 kg). Provided the drone carries a UK2 or equivalent C2 class mark, it operates in the **Near People (A2)** sub-category. The

Component	Weight (g)
Drone (airframe)	1,319
Battery	519
Payload	195
Total (MTOM)	2,033

TABLE II: Component weights and Maximum Take-Off Mass (MTOM)

qualification requirements to operate the drone under this classification are outlined in the following section.

The key rules for drones that come under UK2 or C2 class mark are as follows:

- Maximum takeoff altitude is 120 m above ground level.
 - Keep at least 30m horizontally from uninvolved people, or 5m when using the built-in low-speed mode.
 - Not supposed to fly over uninvolved people intentionally
 - An operator ID, Flyer ID and an A2 Certificate of Competency (A2 CofC) are mandatory.
- ii. **Airspace Restrictions**
According to UK flight rules, drone operations are restricted to a maximum height of 120 m (400 ft). As a UK resident, it is mandatory to comply with the country's drone operation regulations and airspace restrictions. Drone operations are banned within Flight Restriction Zones (FRZs). area around 5km from aerodromes, airports, and heliports. It is essential to get ATC clearance before take-off.
- iii. **Environmental protection**
To protect the geological, ecological and natural environment, specific areas are designated and protected by Natural England, like Sites of Specific Scientific Interest. Enabling inclusion and exclusion zones in the path-planning algorithm enables the system to automatically trigger landing or disarm procedures when a geofenced or restricted area is unknowingly breached. Taking care of the geofence and triggering landing or disarming operations are necessary to protect wildlife according to [7]. The breaching act can be avoided by following the method:
- Path selection: Avoid entering exclusion zones
 - Higher altitude to reduce disturbance to wildlife
 - Reduce noise during flight.
 - Minimize hovering duration.
- iv. **Data Protection And Privacy (GDPR)**
Make sure to comply with UK GDPR rules and the Information Commissioner's Office during drone operations.

It is mandatory to consider and respect individual privacy. The following points are key operational requirements that one needs to take care of:

- Operators must be aware of details like image-capturing resolution, zoom functionality and recording range.
- Have control over start and stop operations during the recording process.
- Avoid capturing uninvolved people in the mission.
- Wisely handle the data storage.

E. Failsafes

Prior to flights the reference parameters were uploaded and retained. These were given as the baseline parameter list upon which the Hexsoons were tested prior to development.

1) Battery

Disabled under the supervision of safety pilot, who monitored the battery state.

2) Telemetry

About the Data Link parameters, we went with default values. The `FS_DR_ENABLE=2` starts an RTL function on loss of telemetry for more than 30 seconds, as `FS_DR_TIMEOUT=30` suggests.

3) Crash and Vibration

The crash check is on as well as the vibration failsafe for minimum safety checks.

4) Extended Kalman Filter (EKF)

Regarding the EKF failsafes, we chose the threshold value for balancing genuine data against noise received by the sensors. The filtering window value represents a fair number which can filter values while remaining able to react quickly to a failure.

5) GCS

`FS_GCS_ENABLE=0`. Disabling this failsafe was done in agreement with the safety pilot, leaving it up to the company pilot and the GCS to decide how to act in situations like this.

6) RC

Upon calibrating the RC, based on the minimum `FS_THR_VALUE` we chose the value 970 as the safe value.

7) Geofence

Lastly, our Geofence failsafes: The initial mindset was that on fence breach the drone would RTL, but before the demonstration we decided to keep things simple and safe, which is why we chose a Land sequence and not an RTL. After being asked about it during the presentation, we decided that for a demonstration of our project the best order of actions would be: Loiter → RTL → Land on geofence breach. `FENCE_TYPE = 7` (Polygon + Circle + Altitude) means three active fence types for a realistic approach, with a maximum altitude of 50m, minimum of -10m (considering floor elevations). Margin of 2m and a total of 13 polygon points,

with no advanced settings enabled as indicated by FENCE_OPTIONS=0.

F. Pilot Strategy

For Fenswood Farm flights and trial days the company had three active roles. The Company pilot operated the wireless remote controller to control the drone. Their role involved controlling the drone for all manual tests and in situations where automation was immature. The Safety Pilot provided by the University of Bristol (UoB) supported the Company Pilot with expertise and intervention during high risk events and incidents. A Mission Planner Ground Control System (MPGCS) operator would conduct callouts informing MAVLink status, Battery Level, Global Positioning System (GPS) fix and confirm the drones telemetry with the active pilot. Lastly, the Automation GCS (AGCS) operator would interact with the Automation Frontend and direct the drone through rescue actions. To compensate for the low state of technology readiness throughout development; the strategy employed for safety during flights was to delegate safety actions to the Company Pilot, Safety Pilot and MPGCS operator. The AGCS has the capacity to perform emergency actions such as Landing, RTL or Emergency Disarm during flight but due to observing unforeseen complications with automation on trials this was considered high risk. Prioritising safety when operating flying vehicles is of the highest importance for this development project.

G. Safety Analysis

The lab environment posed few unique risks as the Hexsoon drone platform was provided pre-assembled with all fittings and UoB common lab Health and Safety guidance was observed for all indoor work. While indoors, no one was permitted to power the drones motors and attach propellers at the same time to reduce collision risk. The Raspberry Pi uses a 15W, 5V/3A power supply and the Cube can be powered using a 5V/1A USB interface to PC. As such there was little electrocution risk. Therefore focus surrounded flight trials.

1) Risk Identification

Drone Fly-away - The drone travels out of the designated flight area. This risk can cause environmental damage, collision risk or interference with Bristol Airport flight paths. Mitigated by enabling geofences around flight area boundaries that force RTL 2m from approach. Should this still occur the pilot toggles to LOITER to trigger the RC Override and immediately lands the drone

GCS Comms Loss - Mission Planner was deemed the trusted software used for observation during trials. This loss inhibits safety critical mission monitoring. This risk is mitigated by keeping attention for freezes or GCS link warnings, with either indicated the GCS operator will call out for the pilot to

land the drone and re-establish connectivity while grounded.

GPS/EKF loss - Loss of GPS satellite contact or GPS fix < 3. 1 = No Fix, 2 = 2D Fix, 3 = 3D Fix, 4 and above indicate Differential GPS with various corrections applied for better positional accuracy. This risk can lead to uncertainty in the drones positioning and collision risks. Mitigated with pre-flight observations of GPS Fix and EKF monitoring. Should the GCS operator observe a GPS fix lower than 3 the flight is immediately grounded for investigation of hardware.

Battery Depletion - Battery unable to sustain predictable or maintain altitude flight due to low battery. Mitigated by ensuring the battery is charged to within 95% of capacity. Should any indications appear the GCS pilot will call out and the Pilot will manually land the drone.

Erratic movement - The drone moving unpredictably or in conflict with commands. Erratic behaviour can cause injury or free-fall. The mitigation for this occurrence is denying any flight with automation software that is untested in simulation. Should this behaviour present during flight the action is for the pilot to set LOITER mode, disable the FSM and land the drone with RC controls.

2) Risk Matrix

For the risk matrices the following key was used to indicate the likeliness of incident occurrence and the severity of that incidents occurrence. These have been subjectively determined. As shown in Table III, probability has been rated on a 5-point scale from very unlikely to very likely, and impact ranking from negligible to severe. As illustrated

Probability (P)	Impact (I)
1 – Very Unlikely	1 – Negligible
2 – Unlikely	2 – Minor
3 – Possible	3 – Moderate
4 – Likely	4 – Significant
5 – Very Likely	5 – Severe

TABLE III: Probability and Impact Legend

in Table IV, risk factor values are calculated by combining probability and impact legend values.

Index	Risk	RF	New RF
A	Drone Fly-away	15	6
B	GCS Comms loss	8	3
C	GPS/EKF loss	4	2
D	Battery Depletion	8	1
E	Erratic movement	9	4

TABLE IV: Risk Factor (RF) index

Table V depicts the risk factors before implementing mitigation strategies, while Table VI shows the

Risk Factor Before Changes					
P\I	1	2	3	4	5
5					
4		B			
3				E	A
2		C		D	
1					

TABLE V: Risk Assessment Matrix prior to implementing mitigation strategies

Risk Factor After Changes					
P\I	1	2	3	4	5
5					
4					
3	B				
2		E	A		
1	D	C			

TABLE VI: Risk Assessment Matrix after implementing mitigation strategies

risk assessment matrix attained after implementing mitigation strategies.

H. Development Strategy

Modularity - The system was split by function into the Vision Block, Automation Block, Abstraction Layer, GUI Frontend and GUI Backend. Within this, we maintained a modular system where scripts could be developed independently. This allowed Vision and Pathing functions to have no dependency on ardupilots simulation and to develop asynchronously. These can then couple with the drones FSM script with state logic pre-defined for integration.

Software In the Loop (SITL) - ArduPilot's autotest tools contain the simvehicle.py [8] script, which when configured can simulate the inputs and outputs of the flight controller. This tool can distribute its output wirelessly using User Datagram Protocol (UDP) to a configurable IP address and Port. With a development Raspberry Pi configured to use MAVProxy and a camera the SITL MAVLink output can be interfaced with the Pi on a local network and then rerouted back to a development laptop. This replicates most of the working system for working off-site. Parameters can be uploaded as well as missions observed through Mission Planner or QGroundcontrol software [9]. A downside is that functionality that does not match presents a challenging debugging task due to ambiguous error messages.

Hardware In The Loop (HWIL) - With the drone directly interfaced to a development computer systems such as the GPS and Servos can be interrogated

and adjusted. This is the closest to direct operation possible while in the lab environment. Development using the drone is limited however, as within the building it is not possible to get a GPS fix and then arm the drone. In practice, SITL development was most practised.

III. SYSTEM DESCRIPTION

A. System Architecture

1) Drone Configuration

SPEED: - 2.5m/s was selected and enabled through the AGCS. The default 5m/s speed introduced instability in the drones flight. However, due to time constraints we were unable to measure change to battery consumption at this setting. This factor would, with more time and testing heavily influence the speed we choose for the mission and variable speeds are considered for different legs of the mission. Such as, 4m/s for travel to the search area, 2.5m/s for the lawn search and 1m/s for the PLB search.

HEIGHT - The height chosen was 20m, trials indicated that the starting height of 10m was too low when approaching the South-Eastern hill at Fenswood farm. Otherwise, this provided a comfortable vision footprint and was the height at which the configured vision detection had the most stable capture of a person Fig. 7.

ROUTE - An example of the mission planning and waypoint configuration is shown in Fig. 3. The path to the flight area is hard coded to ensure compliance with mission constraints, while the waypoint distribution follows the discussed coverage pattern within a hard coded polygon defined by provided mission co-ordinates. The take-off location is positioned near the boundary to minimise transit time to the search region, and the path layout is designed to balance coverage efficiency with operational safety.



Fig. 3: Mission route illustrated in ardupilot Mission Planner software

2) RC Setup

Based on recommendations given by the UoB provided Safety Pilot the RC mode switch was configured to enter LOITER when pushed into the upper and lower positions with neutral setting GUIDED state. This reduces the Pilots mental stack when taking physical control as flicking the switch in ANY direction results in safely regaining control of the Drone from its autonomous systems.

3) System Overview: Drone

Frame: Hexsoon EDU-450 The drone platform given in figure 4 is reliable for research, testing and experimenting purposes, featuring a frame designed with 4 motors, ESCs and propellers [10].

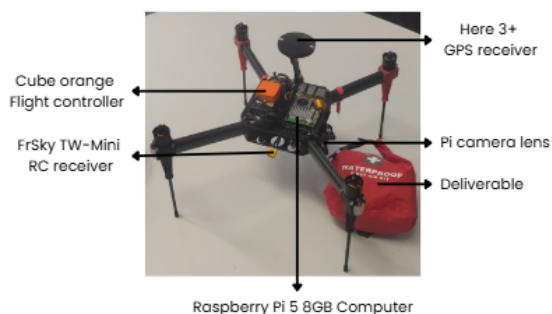


Fig. 4: Hexsoon EDU-450 Frame with RC and GPS receivers indicated

Companion Computer: Raspberry Pi 5 8GB RAM, configured with Raspberry Pi OS (64-bit), powered via a 5V connector from Hexsoon's power distribution. Responsibilities:

- Establishing the WLAN interface for data transfer between the Orange Cube, the Mission Planner GCS (MPGCS) and the Automation GCS (AGCS).
- Interfacing with the drones primary camera
- Performing various Vision detection algorithms
- Hosting the Automation Finite State Machine
- The generation of waypoints and AUTO missions
- Sending pymavlink commands to the Orange Cube
- Logging MAVLink messages and variables within ardupilot STATUS messages

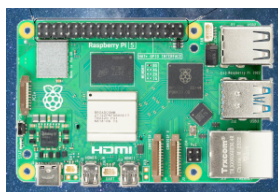


Fig. 5: Raspberry Pi 5

Flight Controller: Orange Cube

- Position and State Estimation

- Stabilisation
- Motor and actuator control
- Mission Execution
- Safety and Failsafes
- Power and System monitoring

AGCS: Multi-script python program with tkinter GUI:

- Video feed
- Button based commands
- .kml parser
- MAVLink Telemetry Display
- Drone FSM display
- Comms/connection indicators
- Object detection indicator
- Speed control
- Command log
- ACK/NACK status display
- PLB and Geofence upload
- Debugging tools
- FSM Safety override clearing

The AGCS GUI was designed to be as simple as possible visually so as not to overwhelm an operator in the case that the primary user delegated by the company is not present. All safety related buttons that may interfere with the regular operation of the drone are located below the visual feed so as not to be accidentally triggered. These are also visually coded such that it is more obvious they are to be used in a unique safety critical situation. In practice due to the fragility of the software and flight safety rules governed by the UoB UAS Ops manual regarding dropped articles 5.2.3 [11]. The company felt it best to disable the ability to engage the emergency motor disarming function as if triggered incorrectly while the drone has directional momentum this could cause a very high safety risk as it falls. As such these controls were left in the care of the Company Pilot and Safety Pilot. Debugging features included a semi-obsured Message Monitor button which indicated the time since the last UDP packet was received from one of the many managed threads. Geofence upload at the time of projects end had not completed testing. Test Square is a debugging and flight trial action to fly in a small defined route for observation. MPGCS: A second computer only running Mission Planner for conducting Ground Station callouts and monitoring the drones telemetry as a safety-critical operation.

Development being conducted on a Linux computer meant that QGroundControl was used for the testing and demonstration of the autonomous functions with SITL and HITL. However, Mission Planner was chosen as the primary GCS as there was more familiarity amongst potential users and allows any company member to participate in the GCS operating role

4) Communications

The system performs data communication between the onboard computer and the ground control station over a network to support video streaming, detection result feedback, and the exchange of control-related information. To meet real-time performance requirements, the communication layer primarily adopts UDP. Compared to the Transmission Control Protocol (TCP), UDP offers lower latency and does not require connection establishment or bidirectional handshaking, thereby reducing waiting time and the risk of blocking. This allows for lower end-to-end latency, which is critical for the perception–control feedback loop in this system, where detection results must be delivered promptly to enable timely control responses. In the implementation, UDP is used to transmit both the video stream and detection-related data. The video stream is transmitted to the ground control station for real-time operator monitoring, while detection outputs and control-related information are transmitted in a lightweight data format to reduce bandwidth usage and improve system responsiveness. In addition, detection outputs are consumed locally by the onboard control logic while simultaneously being transmitted to the ground control station for monitoring and potential system improvement. The use of UDP enables this dual use of data by allowing non-blocking transmission without delaying the onboard control loop. For time-sensitive control information, UDP provides higher transmission efficiency, whereas TCP is generally more suitable for applications requiring higher reliability. However, in this system, since detection and control data are continuously updated, occasional packet loss is acceptable. Furthermore, for live video stream, temporal responsiveness is prioritised over image quality, leading to the choice of UDP to ensure overall real-time performance. UDP does not guarantee reliable data delivery, meaning packets may be lost or arrive out of order. In the current implementation this is mitigated by retransmitting packets and introducing simple acknowledgement mechanisms at the application layer. For this project a local wireless network required for this transmission has been verified to function effectively in open outdoor environment. In other environments this may be affected by unrelated network traffic and such scenarios have not yet been tested. In addition, the current communication is unencrypted, which introduces potential security risks. Although this is acceptable in a controlled environment, future deployment would require improved communication security and more robust transmission strategies.

B. Subsystem Architecture

1) Overview

Multiple scripts are hosted either on the companion computer or Automation GCS itself. The

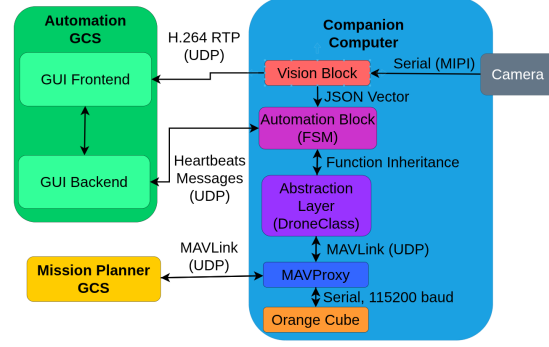


Fig. 6: Simplified representation of System Interfaces

following overview will discuss the functionality across the system blocks and this may indicate behaviours that occur between multiple scripts however where necessary extracts will be used. Primary scripts include Drone_backend.py for the AGCS and PI_State_machine.py (Pi FSM) for the companion computer (RPI). Other scripts were supplemental and in keeping with our modular design ethos, were adaptable should any imported functions or classes changed.

C. Drone Subsystems

1) Automation Block

The automation block includes the Pi FSM. This block manages the high-level mission logic and coordinates drone actions. These outputs are based on GCS commands and camera inputs. In addition, this module facilitates the communication of vision and vector data streams to the Automation GCS.

2) Coverage (Pathing)

Altitude-aware lawnmower coverage planner.

The coverage planner generates a boustrophedon, or lawnmower, path over an arbitrary polygonal search area. Its role is to provide systematic visual coverage during the broad search phase and any later research of a smaller focus area. Unlike a fixed-spacing waypoint generator, the implemented planner explicitly links the path geometry to the camera footprint, so the search pattern scales with altitude rather than being hard-coded for only one operating height. A key design choice in this module is that the image footprint is modelled as approximately linear in altitude for a nadir camera over relatively flat terrain. At a reference altitude of 50 m, the code stores a calibrated footprint width and height. For any commanded search altitude h , the footprint is rescaled by h / h_{ref} , giving a cross-track width $W(h)$ and along-track height $H(h)$. The adjacent-row spacing is then computed as $\Delta y = W(h)(1 - O_{\text{side}})$, where O_{side} is the side-overlap fraction. If intermediate points along a row are required, an along-track step can also be generated as $\Delta x = H(h)(1 - O_{\text{forward}})$. This

means that when the vehicle flies lower, the planner automatically reduces the row spacing to preserve coverage; when it flies higher, the spacing increases to avoid redundant oversampling.

$$W(h) = W_{\text{ref}} \left(\frac{h}{h_{\text{ref}}} \right)$$

$$H(h) = H_{\text{ref}} \left(\frac{h}{h_{\text{ref}}} \right)$$

$$\Delta y = W(h) (1 - O_{\text{side}})$$

$$\Delta x = H(h) (1 - O_{\text{forward}})$$

Once the row spacing has been defined, the planner selects a sweep direction by estimating the principal axis of the search polygon using a simple PCA-based heuristic. The polygon is rotated into this sweep frame so that coverage rows can be constructed as horizontal scan lines. For each row, the planner computes all intersections between the row and the outer boundary, pairs the intersection points into in-polygon intervals, and then subtracts any intervals that lie inside exclusion holes. This subtraction step is important because it allows the vehicle to maintain coverage of the valid search area without generating waypoints that pass through the SSSI or other protected regions.

The remaining intervals are stitched into a boustrophedon sequence. Consecutive rows are traversed in opposite directions to minimise deadheading, while a small boundary margin trims segment endpoints slightly inward to reduce the chance of touching the polygon edge because of GPS or modelling error. The planner can also bias the initial row direction toward a chosen start location, which reduces the transit distance between the aircraft's current position and the first useful search segment.

3) Abstraction Layer

The abstraction layer is a class imported to the Pi FSM that facilitates the more direct commands and calls to the Orange Cube via MAVLink. The decision to use such an abstraction was to simplify the readability of the Pi FSM and reduce the unintended technical debt incurred over time by needing every contributor to have specific knowledge in order to modify the script. The class also uses protected variables for telemetry and utilises locks that disallow any other function from changing these values but permits any external function to read telemetry. This prevents mutex errors.

4) Vision Block

The vision module is responsible for detecting injured personnel on the ground from the live video stream captured by the onboard camera, and providing detection outputs to downstream modules for state transition control.

The input to the vision module is a real-time video stream acquired from the onboard camera. The video stream is processed by a lightweight object detection model deployed on the onboard computer (Raspberry Pi 5). The model uses a YOLO26n deployed using OpenVINO, which is configured to balance detection accuracy and computational efficiency under hardware constraints. In addition, parameters such as image resolution and detection frame rate are also adjusted to optimize the video processing pipeline.



Fig. 7: Example of vision-based detection output showing a detected person with bounding box in the onboard video stream.

The outputs of the vision module include detection results such as bounding boxes and the relative position of the target within the image frame, as illustrated in Fig. 7. The detection video with bounding boxes is transmitted to the ground control station via a network port for operator monitoring. For detection results and control-related data transmission, UDP is used instead of TCP to reduce latency and avoid blocking, thereby meeting the low-latency requirements of the perception-control feedback loop.

In particular, the module computes normalized position offsets (e.g., dx and dy) to represent the relative displacement between the target and the image centre. A lightweight communication approach is used to transmit detection outputs, where structured data (i.e., JSON-formatted position information) is used instead of raw image data. This reduces transmission size, lowers bandwidth usage, and improves system response speed. These offset values are then passed to the flight control logic for further decision-making.

To improve robustness, detection results are evaluated using averaged outputs over a dynamically updated short time window rather than relying on single-frame outputs. In addition, stability conditions are applied to ensure that only consistent detection results can trigger state transitions, thereby reducing the impact of noise and false detections.

The ALIGN logic is activated once a target is detected by the vision module. It uses the received relative position offsets (dx , dy) to generate navigation correction commands in real time, allowing the

UAV to iteratively adjust its position until the target lies within a defined region around the centre of the camera view. This alignment is achieved through iterative position correction based on dx and dy, reducing the offset until the target is centred within a predefined tolerance.

This module serves as a key perception component in the system. By providing consistent and structured detection information, it supports downstream functions such as alignment and landing. At the same time, it acts as an interface between perception and higher-level flight control logic, enabling reliable state transitions based on visual feedback.

5) MAVProxy

MAVProxy is used as a communication layer between the flight controller, the companion computer and the ground control stations. It forwards MAVLink messages, via UDP packets to enable simultaneous access to telemetry and system status by multiple endpoints. In this system this includes the AGCS and Mission Planner. Telemetry transmission therefore does not have to be managed by the FSM. MAVProxy supports real-time monitoring and command exchange, ensuring that both autonomous control processes and operator interfaces receive consistent system information. For this implementation MAVProxy needs to be manually configured on the Raspberry Pi prior to initialising communications. Automating this is possible however, as IP addresses were hardcoded this would require a more nuanced solution to account for changing computers used as the MPGCS or AGCS.

6) Finite State Machine

The Finite State Machine (FSM) constrains autonomous actions to appropriate situations. The drone begins grounded in the INIT state. INIT resets variables and refreshes overrides. A condition in code checks whether the drone is airborne >1m when the FSM is restarted triggering FLIGHT as the first state. Waypoint missions generated by the FSM are uploaded to the Flight Controller via the abstraction layer.

State	Description
INIT	Initial default state
FLIGHT	Drone stable in air
LAWN	Perform search pattern
AOI	Travel to Area of Interest
RETURN	Return to landing area
PLB	Perform focused search
ALIGN	Center camera on target
DESCEND	Automated descent
LAND	Land at current position
EMERGENCY	Disable state machine

TABLE VII: FSM States, Area of Interest (AOI), Personal Locator Beacon (PLB)

The use of 'AUTO' missions forgoes the need to track telemetry at a high data rate and flag

when each waypoint is reached as this is done by ardupilot reducing the need for more background monitoring threads. GUIDED mode then forms an 'Idle' pseudostate with AUTO being the trigger to start each mission. This allows missions to be interrupted easily by toggling GUIDED, clearing the previous mission, then repeating the process. This is shown in Fig. 8. To illustrate the state transitions the pseudostate 'FREE' indicates to the user that they can freely interrupt the drone with any compatible instruction from the list of 'lawn', 'aoi', 'plb_start' and 'return'. This is because there are two representations of the drones activity considered important for the user. What the FSM is instructing the drone to do and whether the FSM perceives the drone as grounded or flying. Due to the late development of ALIGN and DESCEND states, payload release was not yet implemented

State Transitions (FSM inputs in lowercase)

State INIT:

on 'takeoff' → FLIGHT
on 'init'/reset → INIT
on 'emergency' → EMERGENCY
on 'rtl' → INIT
on 'takeoff_complete' → FLIGHT
on 'airborne_recovery' → FLIGHT

State EMERGENCY:

on 'override_clear' (ground) → INIT
on 'override_clear' (airborne) → FLIGHT

State FLIGHT:

on 'lawn' → LAWN
on 'aoi' → AOI
on 'plb_start' → PLB
on 'return' → RETURN
LAWN on mission_complete → 'FLIGHT'
AOI on mission_complete → 'FLIGHT'
PLB on mission_complete → 'FLIGHT'
RETURN on mission_complete → 'FLIGHT'

ALIGN / DESCEND transitions:

FLIGHT on 'target_stable' → ALIGN
ALIGN on 'centered' → DESCEND
ALIGN on 'lost_target' → FLIGHT
DESCEND on 'descend_complete' → FLIGHT
DESCEND on 'descend_interrupt' → FLIGHT

Landing logic:

FLIGHT on 'landing_event' → LAND
LAND on 'landed' → INIT
FLIGHT on 'land_command' → INIT

7) GUI / Frontend

The following passage describes elements of the gui shown in Fig. 9.

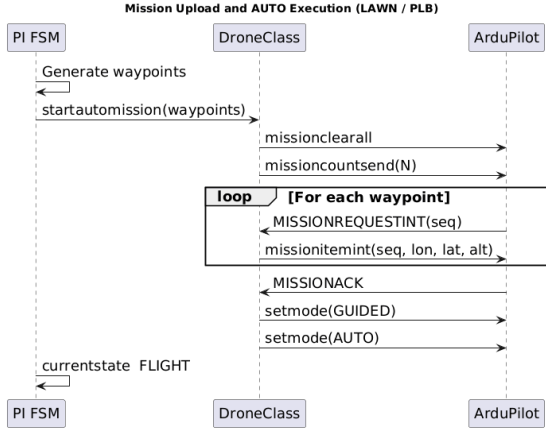


Fig. 8: Sequence diagram illustrating waypoint mission generation and execution for LAWN and PLB states

- 1) **Drone controls** - This section of the GUI indicates 'CONNECTED' for the Connection status and mavlink status for each provided that they independently receive UDP messages within a 3s timeout. The final indicator displays the current State that the drone has currently entered.
- 2) **Upload controls** - This section of the GUI allows for the user to upload a PLB polygon with at least 3 waypoints and initialises the PLB search state via button press. Upon pressing 'Browse KML' the operator can find and upload the .kml file. These can be previewed for confirmation. Below are controls for uploading a Geofence via .csv however this remained untested due to time constraints. Lastly, speed controls for the lateral motion of the drone.
- 3) **Command Log** - Display of communications between the FSM and AGCS indicating the message sent and the Acknowledgement (ACK) received. Here, will display "Pi:" followed by a message. For example: "Pi: FSM - Mode set to GUIDED State=INIT Event=init" Below is an ACK status bar which holds in memory the last received ACK from the Pi FSM
- 4) **Telemetry** - Display of the drone telemetry from MAVLink including: Mode, Arm state, Latitude, Longitude, Altitude, GPSFix, Heading and Ground Speed. This persists even if comms are lost with the FSM. Below the detection indicator is displayed should a person be seen in the live feed.
- 5) **Video Feed** - This displays the direct feed from the drones camera, via UDP H264 video stream. The IP address and Port are configured by the Operator. This facilitated easier bench testing with external camera systems. Videos

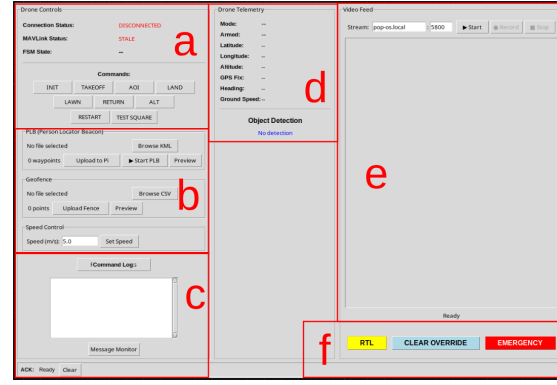


Fig. 9: Automation GUI with boxes indicating areas of interest

can be recorded by button press and saved with date and time.

- 6) **Overrides** - The drone overrides to set RTL and Emergency states exist here, in bright colours to ensure they are apparent as safety controls. These were disabled during trial and demo flights. Clear Override resets the FSM if LOITER mode was triggered.

8) Backend threads

The **pi_status_listener** thread is a free-running thread that waits for FSM messages, filtering heartbeats and instructions for various exit points in the AGCS Front end code. These status messages received over UDP Port 9051. An ACK for every received message is then sent alongside the message and type of message. The responses indicate that there is still an interface between the AGCS and FSM.

The **mavlink_listener** thread reads and parses all telemetry data from heartbeat messages every 0.1s. These messages from the Orange Cube are routed through the Companion Computer via MAVProxy. Messages in this pipeline are used to update the connection status to MAVLink in the Frontend and populate the drone properties in the telemetry section of the GUI.

The **vector_listener** receives JSON UDP messages describing the vector between center of the camera to the detected target. These are then distributed to the Frontend but unused.

IV. REQUIREMENTS VERIFICATION

A. Requirements Summary Table (R01-R12)

B. R01 — Flight Area Containment

Requirement met. The company set geofence, which derived from CAA regulations [6] and also from the constraints of the assignment regarding the Fenswood Farm area. it was tested, while developing in simulation the lawnmower search pattern, using a kml file, to make sure we move on to HITL test safely. During testing the Drone never went outside the Flight Area.

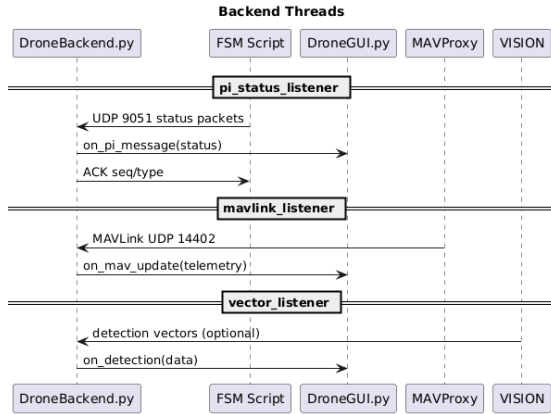


Fig. 10: GUI Backend thread sequence diagram indicating data sources

C. R02 — SSSI Avoidance

Requirement met. Similarly, when testing the SSSI geofence in simulation, as with R01.

D. R03 — Takeoff Location

Requirement met. During every takeoff, the drone was placed under supervision of the safety pilot at $51^{\circ}25'24.2''N$ $2^{\circ}40'17.4''W$.

E. R04 — Maximum Altitude

Requirement met. The company set maximum height in Mission Planner at 50m, as well as in every function which impacted the altitude.

F. R05 — Search and Detection

Requirement partially met. The computer vision model was trained to detect only humanoid figures. Due to time constraints, the model was not configured to identify items of interest.

G. R06 — PLB Response

Requirement met. At the final stages of code integration a PLB function was created.

H. R07 — Landing Proximity

Requirement not met in automation. Due to time constraints there was not enough time to perfect the alignment function, resulting in a manual contingency which meets the requirement. Tested by the company pilot.

I. R08 — Autonomy Level and Interface

Requirement met. The autonomy levels were agreed on in various company meetings and also they were approved by the safety pilot.

J. R09 — RTL and Motor Cutoff Availability

Requirement initially met. The RTL and Motor Cutoff functionality was integrated in both MPGCS, AGCS and RC. After discussion with safety pilot, it was decided to remove the Motor Cutoff functionality from the AGCS for safety reasons; the existence of such a high risk function could accidentally compromise the mission.

K. R10 — Positional and Image Reporting

Requirement not met due to time constraints. Although the model detects target with telemetry, a manual notation would be required with the current implementation.

L. R11 — Designated Company Pilot

Requirement met. Roles assigned before trial flights started.

M. R12 — Software and Log Delivery (GitHub)

Requirement met.

GitHub link: - <https://github.com/argi4s/Group-Project-GreenCo>.

V. EVALUATION

A. Technical Performance Review (Plus/Delta)

The primary technical challenges centred on FSM responsiveness and UDP networking. FSM freezing during Flight Day 3 was traced to a queued MAVLink command conflict, resolved by implementing message filtering by system and component ID. UDP packet loss caused intermittent video feed dropouts, mitigated by application-layer retransmission, though a hybrid UDP/TCP scheme remains a future improvement. Key successes included the altitude-aware lawnmower planner correctly scaling waypoint spacing with operating height, and the YOLO/OpenVINO pipeline achieving stable humanoid detection at 20m altitude as confirmed in flight recordings.

1) Automation Performance

Automation validation was conducted via the validation document and experimentation defined in appendix B 17. In the appendix, the Systems Requirements Table summarises what autonomy level we reached in every aspect of the project and how it was tested. The levels of autonomy demonstrated outside of simulation are: F2 for partial flying autonomy which includes: F2a, F2b, F2c, F2d for execution of predefined action upon loss of communications, EKF, insufficient battery level and geofence boundary approach, respectively. The team achieved S2 level of search and detection autonomy with S2a requiring human confirmation for detection tasks and S2b for localising a detected person of interest. As for the payload, the team achieved a level of autonomy of P1 with P1a providing a payload release function initiated by Pilot/GCS command.

drifted towards waypoint 1, as observed in Figure 13. The safety pilot executed the emergency braking procedure followed by the power cut command, resulting in a crash landing and structural damage. Damaged parts were replaced before the next flight.

4) Flight day 4:

Flight day 4 focused on debugging and resolving the issues encountered during the previous flight. Basic commands including takeoff and landing executed successfully. However, waypoint 1 execution failed: review of the mission logic identified that the FSM was not transitioning out of takeoff due to a measured and expected altitude mismatch, preventing the FSM from entering 'FLIGHT' and blocking AOI. Following correction, the Area of Interest (AOI) navigation command executed as intended in the second test.

B. Environmental and Societal Impact

Building on the cost and operational advantages outlined above, drone-based SAR systems also carry meaningful environmental and societal benefits. The reduced reliance on helicopter deployments, each of which consumes significant quantities of aviation fuel, represents an important step toward more sustainable emergency response operations. As electrically-powered drone platforms become operationally established, they could progressively replace or reduce helicopter hours in the early search and localisation phases, lowering the overall carbon footprint of SAR missions without compromising their effectiveness.

From a societal perspective, this shift is equally significant. Drone-based systems lower the barrier to entry for rescue operations, enabling smaller organisations, volunteer groups and under-resourced agencies to conduct effective search missions that would previously have required expensive aviation assets and highly specialised personnel. Furthermore, faster localisation of missing individuals, enabled by autonomous aerial coverage and onboard perception systems, directly translates to improved survival outcomes, as time is often the critical factor in SAR scenarios.

C. Future Works

The current system demonstrates a functional UAV-based search and detection platform, integrating path planning, vision-based detection, and finite state machine control to achieve core mission objectives.

In terms of system autonomy, the current implementation follows a segmented mission structure, where each stage is executed automatically but requires manual triggering from the ground control station. Future work should focus on enabling automatic transitions between mission phases to reduce

operator intervention, including further validation of modules such as ALIGN under real flight conditions.

From a communication perspective, the current UDP-based approach is susceptible to packet loss and instability in real-world networks. Future improvements could explore more reliable strategies, such as hybrid UDP/TCP schemes, to better balance latency and reliability.

Additionally, more extensive real-world testing is required to evaluate performance under different weather conditions, terrain types, and disturbances, ensuring system reliability and safety.

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VI. APPENDIX

Project Gantt Chart

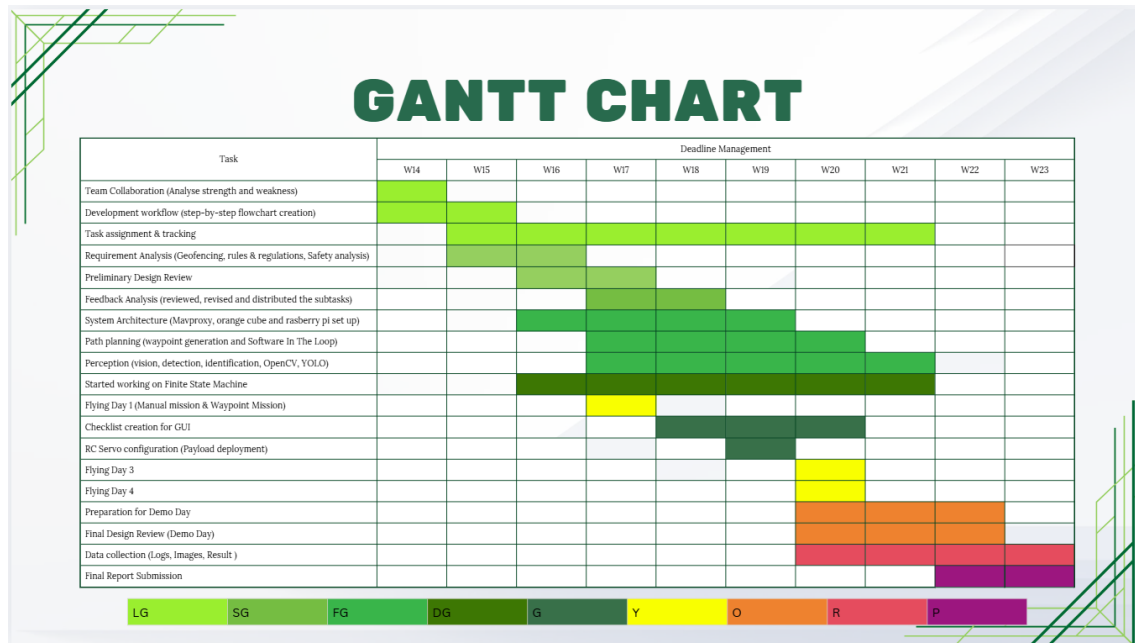


Fig. 14: Gantt Chart

Project Risk Management

ID	Date Raised	Risk Description	Likelihood of the risk occurring	Impact if the risk occurs	Severity	Owner	Mitigating Action
Communication & Coordination							
1	W14	People working on the same system without coordination can lead to integration problems.	Medium	High	High	Team Lead	Establish regular inter-subteam sync meetings and use shared documentation to track integration points
2	W14	Team decisions made unilaterally by a single member without team awareness	Low	Medium	Medium	Team Lead	Define a lightweight decision log; major decisions require acknowledgement from all sub-team leads
3	W14	Language barriers causing misunderstandings in technical discussions.	Low	Medium	Medium	Team Lead	Encourage written documentation of key decisions in agreed common language; use diagrams where possible
Task & Workload Management							
4	W14	Uneven workload distribution across team members.	Medium	Medium	Medium	Project Manager	Review task allocation weekly using the tracking tool; redistribute tasks proactively when imbalance is detected.
5	W15	Delays caused by poorly defined task dependencies causing bottlenecks.	Medium	High	High	Project Manager	Map task dependencies explicitly in the Gantt chart from the outset; flag blocked tasks in weekly standups.
6	W15	Scope creep on minor details causing loss of focus on the MVP and poor deadline management.	Medium	High	High	Team Lead	Define and freeze MVP scope early; any additions require explicit team approval and timeline reassessment.
Personnel Risk							
7	W14	Team member unavailability due to illness or personal circumstances	Medium	Medium	Medium	Team Lead	Ensure no task is held by a single person; maintain handover notes for all active work streams.
8	W14	Loss of a key member holding critical knowledge with no backup.	Low	High	High	Team Lead	Enforce knowledge sharing sessions and written documentation for all critical subsystems.
9	W14	Skill gaps not identified early leading to unrealistic task expectations.	Low	Medium	Medium	Project Manager	Conduct a skills audit at project start; assign training or pair members with complementary skills early.
Planning							
10	W14	Poor time estimation causing milestones to be missed.	Medium	High	High	Project Manager	Apply conservative time buffers in the Gantt chart; reassess estimates after each flying day.
11	W19	Late preparation for trial flights resulting in insufficient time to address issues before Demo Day	Medium	High	High	Team Lead	Schedule trial flights early (W17-W19) with post-flight review sessions and a formal Demo Day readiness checklist.
Decision Making							
12	W15	Delayed on-field decisions causing knock-on delays to parallel development workstreams.	Medium	Medium	Medium	Team Lead	Designate a field decision authority for flying days; empower sub team leads to make bounded decisions autonomously

Fig. 15: Project Risk Management Table

//	Flying Autonomy (Flight handling & mission execution)	Search Autonomy (Sensing, detection & path adaptation)	Payload Autonomy (Deployment, recovery & interaction)
	F	S	P
0	Fully manual piloting with throttle control. Human controls altitude, speed, navigation and mission decisions without an automated control system	Human visually searches via live video. No onboard detection or pre-planning waypoints.	Human manually releases or retrieves payload.
1	Basic flight assistance: stabilization, altitude hold, heading hold.	Camera aids only such as zoom or gimbal stabilization. Human identifies target.	Assisted payload release (single-button drop). Human confirms timing and location.
2	Partial automation: waypoint flight, auto take-off/landing, RTH, obstacle avoidance.	Automated detection indicates person or points of interest. Human confirms identification.	Automated landing or drop routine. Human authorizes release or pickup.
3	Conditional autonomy: drone executes full mission under pre-defined conditions human can take over as required	Onboard vision autonomously detects and returns position of person. Autonomously adjusts path while human supervises.	Drone autonomously lands, stabilises, and deploys defib. Human supervises and intervenes.
4	High autonomy in constrained environments (BVLOS, geofenced). No pilot intervention expected during mission.	Fully autonomous search, identification, coordinate generation, and path re-planning within mission scope	Fully autonomous defib. delivery or retrieval: human chooses landing or drop method, ensures safe placement and confirms release.
5	Full autonomy in all environments and conditions. No human supervision required.	Autonomous search and identification in all environments. Drone learns and adapts without predefined limits.	End-to-end autonomous payload handling: safe landing, human aware interaction, release or recovery, and departure without oversight.

Fig. 16: Automation Definitions inspired by SAE International

Req ID	Autonomy Level	System Requirement	Trigger / Condition	Expected System Response	Acceptance Test	Result (Y, N, P, U, O)	Notes
F0a	0	The drone shall immediately terminate all motors upon pilot or GCS emergency intervention.	Trigger sent MAVLink, Python or RC	Motor thrust reduced to zero within $\leq X$ ms	HWIL: Inject emergency MAVLink / RC signal and verify motor coasts to stop SWIL/HWIL: Simulate release of throttle input; verify altitude maintains at setpoint. Flight: Release throttle during stable hover and observe altitude deviation over time	O	Unimplemented as unsafe in early development software. However, tested with SWIL and functionality is available through disabled code
F1a	1	The drone shall maintain altitude within $\pm X$ m without pilot throttle input when idle.	Throttle input released in flight	Altitude maintained for $\geq T$ seconds	SWIL/HWIL: Inject zero throttle input; verify attitude control outputs remain within limits. Flight: Release throttle and measure roll/pitch/yaw stability over ≥ 15 seconds	Y	Default Ardupilot behaviour in GUIDED and LOITER modes Documented on Flying day 1
F1b	1	The drone shall maintain stable attitude without pilot throttle control.	No throttle input during flight	Roll, pitch, yaw remain within defined limits	SWIL/HWIL: Simulate GCS link loss and verify correct safe action state transition. Flight: Disable GCS link during autonomous flight and confirm safe action is executed	Y	Demonstrated on Flying Day 1, 2, 3, 4 when drone configured to GUIDED or ALT HOLD modes
F2a	2	The drone shall execute a predefined safe action upon loss of communication with the GCS during autonomous flight.	Loss of GCS link for $> T$ seconds	Safe action (e.g. RTL / Land / Hover) initiated within $\leq Y$ seconds	SWIL/HWIL: Inject EKF/GPS fault or disable GPS and observe safe action trigger. Flight: Induce GPS degradation (controlled environment) and verify autonomous navigation terminates safely	U	Unimplemented, development software has frequent dropouts due to UDP comms. MP GSC loss monitored by Company Pilot.
F2b	2	The drone shall execute a predefined safe action upon detection of global positioning error (EKF error) exceeding defined limits.	GPS error exceeds threshold (EKF error raised)	Safe action initiated and autonomous navigation ends	SWIL/HWIL: Force battery level / fault flag and verify safe action selection. Flight: Fly until low-battery threshold or inject fault and confirm safe action is executed	P	Implemented, Untested. Drone behaviour set to RTL when EKF unhealthy failsafe encountered by Ardupilot
F2c	2	The drone shall execute a predefined safe action upon low battery flag or system fault during autonomous flight.	Battery below critical threshold or fault detected	Safe action initiated within ≤ 2 seconds	SWIL/HWIL/Flight: Command trajectory toward geofence and verify boundary response logic.	U	Unimplemented, the Ardupilot mechanism exists and operates in the same manner as EKF, FENCE and other configured failsafe actions.
F2d	2	The drone shall execute a predefined response when approaching a geofence boundary during autonomous flight.	Distance to geofence ≤ 2 m	Boundary response (stop / reroute / RTL) executed	SWIL/HWIL/Flight: Command trajectory toward SSSI and verify boundary response logic.	P	SWIL: Drone refuses waypoint missions outside of Fence area. Geofence configured to RTL when 2m from fence. Flight: Untested during due to time constraints.
F3a	3	The drone shall prevent entry into SSSI designated airspace during autonomous flight.	Planned or active trajectory crosses into SSSI	Drone does not enter restricted airspace	SWIL/HWIL/Flight: Command trajectory toward SSSI and verify boundary response logic.	P	As above, configured SSSI as excluded by fence
S0a	0	The system shall provide a live video feed to the operator.	Camera active	Live video displayed	HWIL: Connect camera to system and verify live feed. Flight: Verify live feed during manual flight	Y	HWIL: Drone bench tested to capture live video feed. Flight Drone recording of flights captured drone_recording_20260331_151917.avi
S1a	1	The system shall provide camera assistance functions without automated target detection. The system shall automatically detect potential persons or points of interest and notify the operator.	Camera aid enabled	Video enhanced only	HWIL: Calibrate and verify image post processing on bench such as colour correction / stabilisation. Flight: Verify aids operate in flight without False Positive alerts	Y	HWIL: Colour correction confirmed, Flight Colour corrected captures recorded drone_recording_20260331_151917.avi
S2a	2	The system shall require human confirmation before classifying detected objects as valid targets.	Target appears in sensor view	Detection alert generated	HWIL/Flight: Present visual stimulus / live person to camera and verify detection trigger or test detection from live feed during flight	Y	HWIL: Detection demonstrated, Flight Detection captured in recording – requires parameter reconfiguration drone_recording_20260331_152508.avi
S2b	2	The system shall autonomously detect human as person of interest and output estimated target position.	Detection alert issued	No classification without confirmation	HWIL: Trigger detection during 'flight' present Pilot/GCS instruction to confirm or deny True Positive target	U	Detection confirmation unimplemented due to time constraints however, Automation GCS operator can intervene or ignore false detections at will.
S3a	3	The system shall autonomously detect human as person of interest and output estimated target position.	Person detected	Coordinates generated	HWIL: Trigger detection during 'flight' and return drones current GPS position.	U	GPS position unimplemented due to time constraints
S3b	3	The system shall autonomously adjust its position to approach person of interest in preparation for next mission step	Drone travels to hover above visual target	Drone travels to hover above visual target	SWIL/HWIL/Flight: Move image across camera and verify if tracking commands issued in GCS or observe drone tracking during flight	N	Unintended automation of alignment upon descent in development
S4a	4	The system shall autonomously conduct search and detection within a predefined operational area without pilot intervention.	Target motion detected	Autonomous search executed	SWIL/HWIL/Flight: Run mission logic in SWIL and verify search logic executes.	P	SWIL: Traced to search area and coverage pattern in search area. This operation is conducted with live video feed and indicates targets but no further functionality
S4b	4	The system shall autonomously adjust its coverage pattern based on detections or updating external information	Mission start	Tracking maintained	SWIL/HWIL/Flight: Execute mission in geofenced area and verify mission logic executes as expected and adjusts pathing when supplanted with PLB co-ordinates	P	SWIL: As indicated above. PLB co-ordinates can be uploaded mid-mission for secondary coverage pattern.
P0a	0	The system shall allow the human operator to manually release or retrieve the payload.	Human on the ground opens payload mechanism	Payload released or retrieved only by direct human action	Bench: Manually release payload	Y	While not recommended, one CAN manually ply open the actuated payload hook
P1a	1	The system shall provide a payload release function initiated by Pilot / GCS command	Pilot / GCS issues payload release command	Human authorizes landing for payload operation	Bench (HWIL): Send payload release command. Verify payload actuator triggers once, confirm correct action and timing	Y	Configurable as WINCH or SERVO14. Payload release can be actuated by RC and python. PWM values configured as 1050, 1550 and 2050
P2a	2	The system shall execute an automated landing routine upon human authorization.	Human authorizes landing for payload operation	Drone payload routine begins upon GCS instruction	SWIL/HWIL/Flight: Verify automated landing. Manually trigger payload release command.	Y	Demonstrated on Flying day 2 on Flying day
P3a	3	The system shall autonomously land, stabilize, and deploy the payload while human supervises and intervenes	Mission reaches payload deployment phase	Stable landing and controlled payload deployment	Bench (SWIL/HWIL): Simulate autonomous payload, test user intervention. Flight: Execute autonomous deployment and attempt to interrupt	N	Autonomous deployment logic unimplemented

Fig. 17: Systems Requirement Table

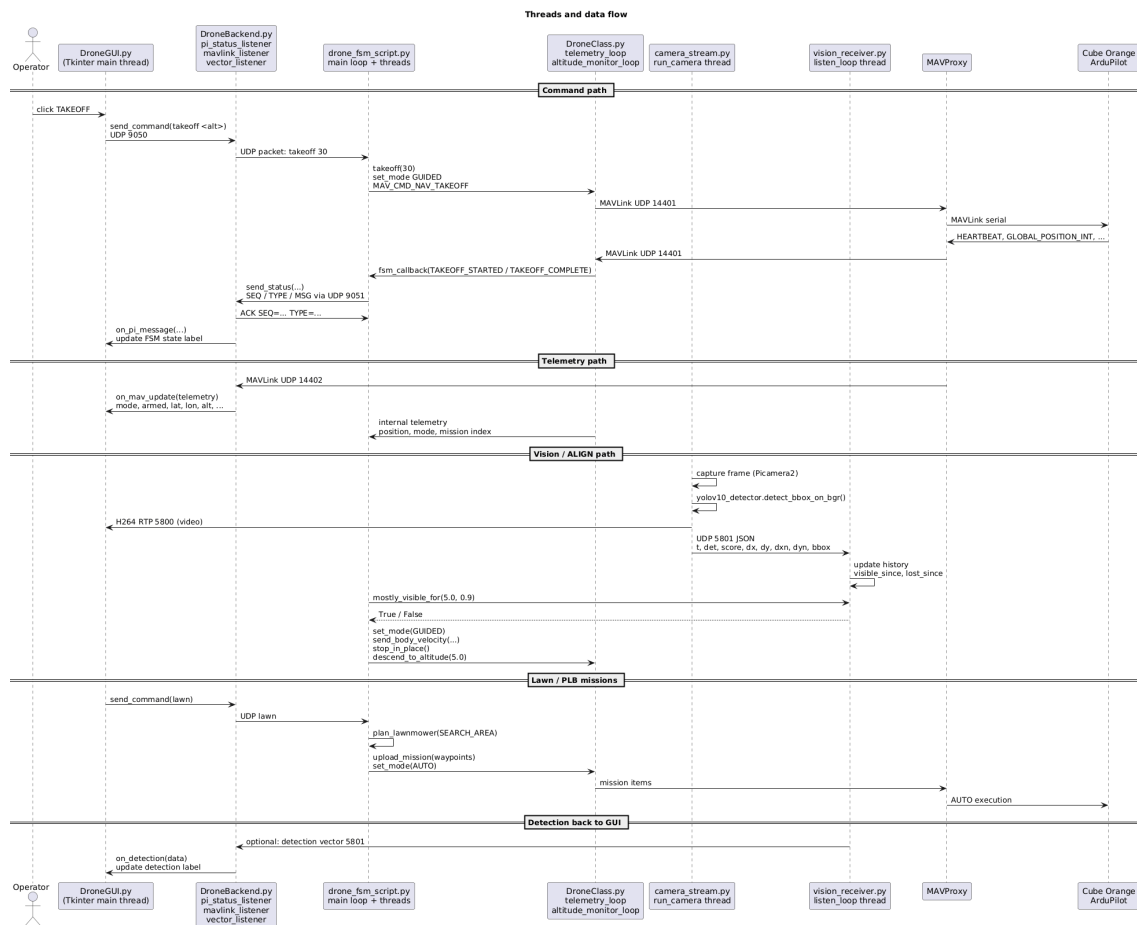


Fig. 18: Sequence diagram displaying every thread in the automation system

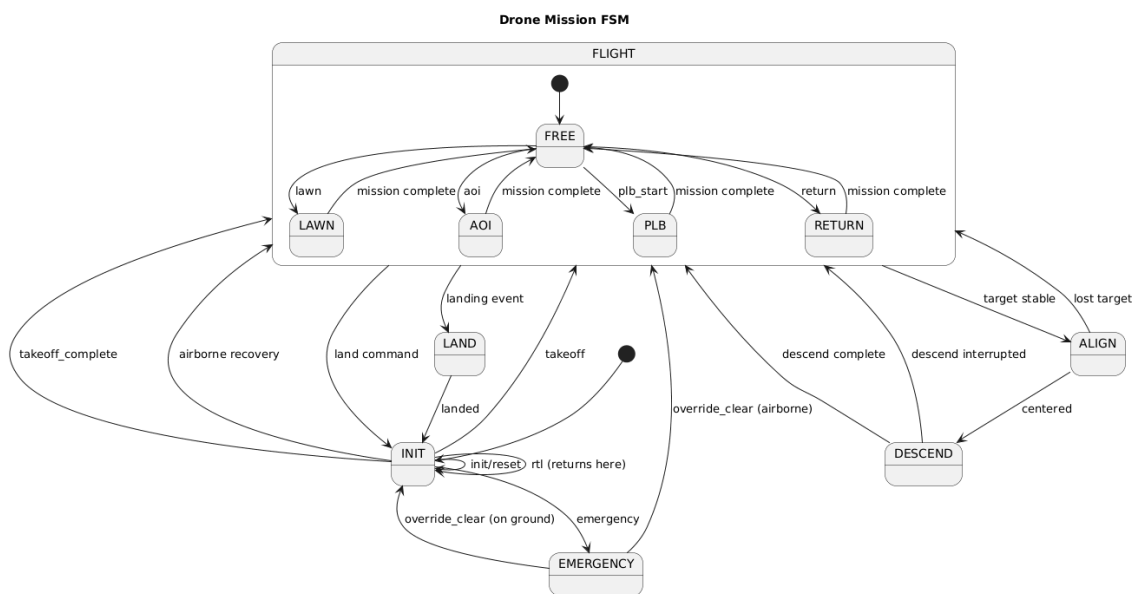


Fig. 19: Top Level state diagram indicating entry points when grounded and airborne